

Reviewer Roadmap

Locked Gravitational Saturation Cosmology, Version 1.1

Purpose

This roadmap allows a specialist to evaluate the portions of the 417-page technical manuscript most relevant to their expertise without reading it sequentially from page one. The full manuscript remains the authoritative source.

Fast orientation

Reviewer interest	Primary sections	What to check
Physical premise and scope	Preface; Chapters 2-5; Appendix E	Whether the effective-model status, assumptions, and claim boundaries are stated clearly.
Bounce and background dynamics	Chapters 5-6; Appendix A; Appendix D	Modified Friedmann constraint, regular evolution through $H=0$, energy conservation, low-density limit, normalization.
Boltzmann implementation	Chapter 7; Appendix B	How the locked expansion history and canonical perturbations are inserted into CAMB; reproducibility of code changes.
Primary CMB and acoustic scale	Chapters 8-9; Appendix C	Peak structure, 100 θ_{*} , drag horizon, reference model, and parameter locking.
BAO and distances	Chapter 10; Appendix C	Distance definitions, sound-horizon use, comparison method, and covariance handling.
Growth and S_8	Chapter 11; Appendix C	Growth equation, normalization, σ_8 and S_8 calculation, and whether perturbations are fully included.
CMB lensing	Chapter 12; Appendix C	Window functions, covariance, chi-square construction, control model, and interpretation.
Robustness and falsifiability	Chapter 13; Appendix E	Parameter sensitivity, no-retuning rule, failure conditions, and what future data could rule out the model.
Complete equation and dependency audit	Appendices A, D, and F	Symbol consistency, equation dependencies, units, assumptions, and cross-references.

Requested review questions

1. Are the equations internally consistent under the stated normalization and assumptions?
2. Does the contraction-era transfer conserve total energy as written?
3. Is the limiting treatment at $H = 0$ mathematically regular and numerically implemented without division by H ?

4. Are the scalar perturbation equations and stress-energy sources implemented consistently in the Boltzmann hierarchy?
5. Can the quoted numerical results be independently reproduced from the released materials?
6. Are the observational comparisons statistically sufficient for the claims made?
7. Which claims should be strengthened, weakened, or separated into future work?

Important claim boundary

The manuscript presents an effective cosmological realization. It does not claim that the saturation potential or the high-density modified Friedmann constraint has been uniquely derived from a complete microscopic theory of gravity. Stability statements apply to the canonical scalar-field sector analyzed in the manuscript.